DATA SCIENCE MINOR PROJECT REPORT

PROJECT REPORT

LOVELY PROFESSIONAL UNIVERSITY

Transforming Education Transforming India

(Project Semester January-April 2025)

Exploratory Data Analysis of Stock Market

Submitted by: Hrishikesh S

Registration No: 12406610

Roll No: 08

Section: 224YM

Course Code: INT 375

Under the Guidance of Dr. Baljinder Kaur (UID: 27952)

CERTIFICATE

This is to certify that **Hrishikesh S** bearing Registration no. **12406610** has completed **INT 375** project titled, **"Exploratory Data Analysis of Stock Market"** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort, and study.

School Of Computer Science and

Engineering Lovely Professional University

Phagwara, Punjab.

Date: 21-04-26

DECLARATION

I, Hrishikesh S, student of P132 under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 21-04-26 Signature

Registration No. 12406610 Hrishikesh S

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my faculty mentor for their guidance and support throughout this project. I also thank Lovely Professional University for providing the resources required to complete this project successfully.

TABLE OF CONTENTS

1. Introduction

2. Source of Dataset

3. Exploring the Dataset

4. Data Preprocessing

5. EDA project

6. Analysis on Dataset

7. Conclusion

8. Future Scope

9. References

1. Introduction:

The stock market is a dynamic and complex system where prices of stocks fluctuate based on various factors such as company performance, investor sentiment, and global economic conditions. With the rapid growth of data availability, analyzing stock market data has become an important application of data science.

This project focuses on performing Exploratory Data Analysis (EDA) on historical stock market data to understand patterns, trends, and relationships between different variables such as opening price, closing price, high, low, and trading volume. The primary objective of this project is to extract meaningful insights from raw stock data and visualize them in an understandable manner.

The dataset used in this project is obtained from reliable financial sources such as NSE Nifty using Python libraries. Various data analysis techniques are applied, including data cleaning, handling missing values, and transforming data into a structured format suitable for

analysis.

Using powerful Python libraries like Pandas, NumPy, Matplotlib, and Seaborn, the project visualizes stock price movements through line graphs, bar charts, and heatmaps. These visualizations help in identifying trends, volatility, and correlations among different stock attributes.
In addition to EDA, a basic Linear Regression model is implemented to perform numeric prediction of stock prices based on historical data. This provides a foundational understanding of how machine learning can be applied to financial datasets, although stock price prediction remains inherently uncertain due to market volatility.
Overall, this project demonstrates how data science techniques can be effectively used to analyze stock market data, gain insights, and support decision-making. It serves as a practical application of EDA concepts and lays the groundwork for more advanced predictive modeling in the future.

2. Source of Dataset:

- Sources: https://www.kaggle.com

3.Exploring the Dataset:

The first step in analyzing this dataset is understanding its structure and the meaning of each attribute. The dataset contains historical stock market data with the following features:

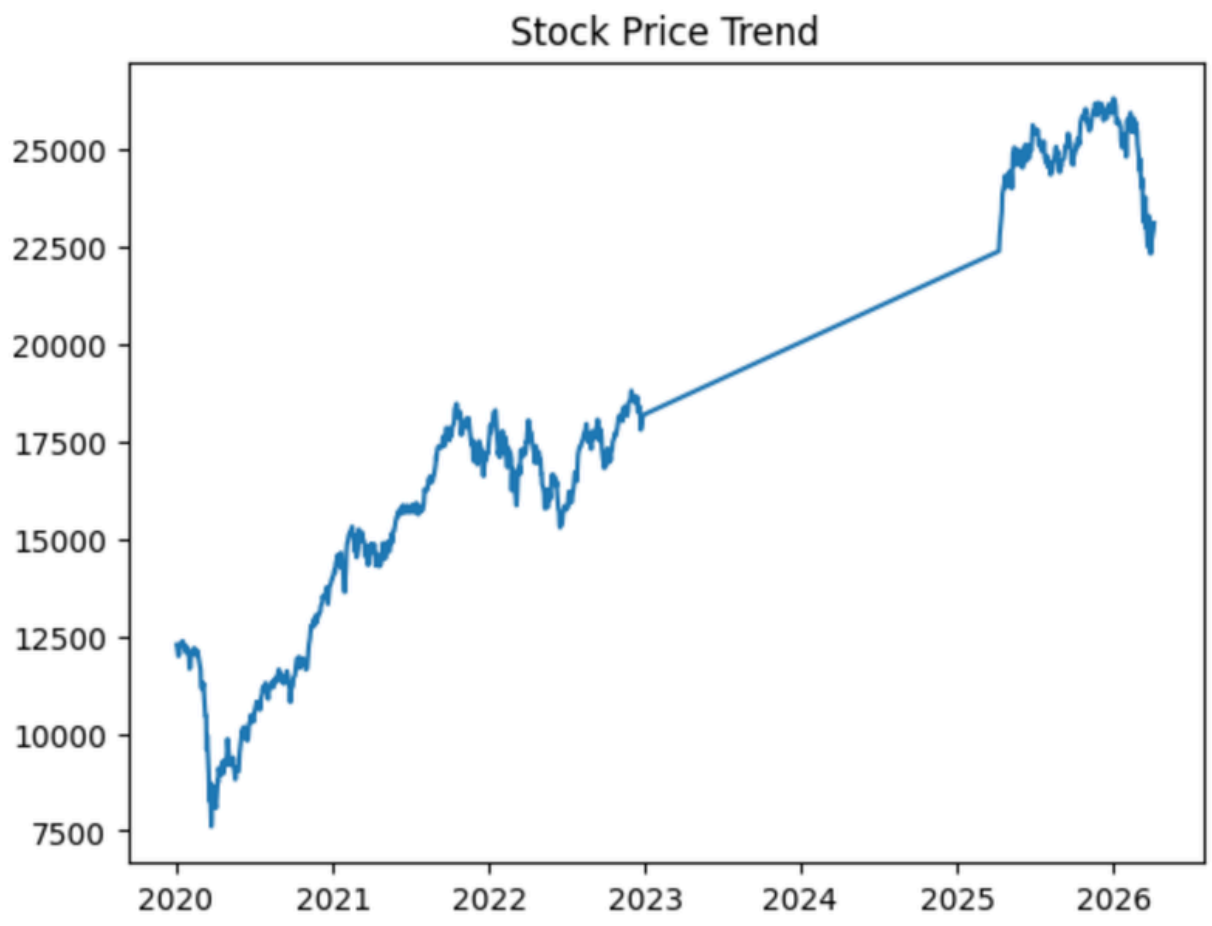
- **Date:** Represents the specific trading day of the stock.
- **Open price :** The price at which the stock started trading at the beginning of the day.
- **Close price :** The final price of the stock at the end of the trading day.
- **High price :** The highest price reached by the stock during the day.
- **Low price :** The lowest price recorded during the trading session.
- **Adj Close (Adjusted Close) :** The adjusted closing price that accounts for dividends, stock splits, and other corporate actions.
- **Volume :** Indicates the total number of shares traded during the day.
- **Daily return :** Represents the percentage change in stock price from opening to closing.
- **Moving Average (MA) :** Shows the average stock price over a specific period (e.g., 7-day, 30-day), helping to identify trends.
- **Volatility:** Measures the variation in stock prices over time, indicating risk levels.
- **Stock Symbol:** Identifies the company (e.g., AAPL, TSLA, RELIANCE).
- **Month:** Extracted from the date to analyze monthly trends.
- **Year :** Helps in analyzing long-term performance of the stock.

4. Data Preprocessing

Data processing involved cleaning and preparing the stock market dataset for analysis. Missing values were handled, duplicates were removed, and the Date column was converted into datetime format. Additional features like daily returns and moving averages were created to enhance analysis and ensure accurate results.

5. Stock Price Trend Analysis

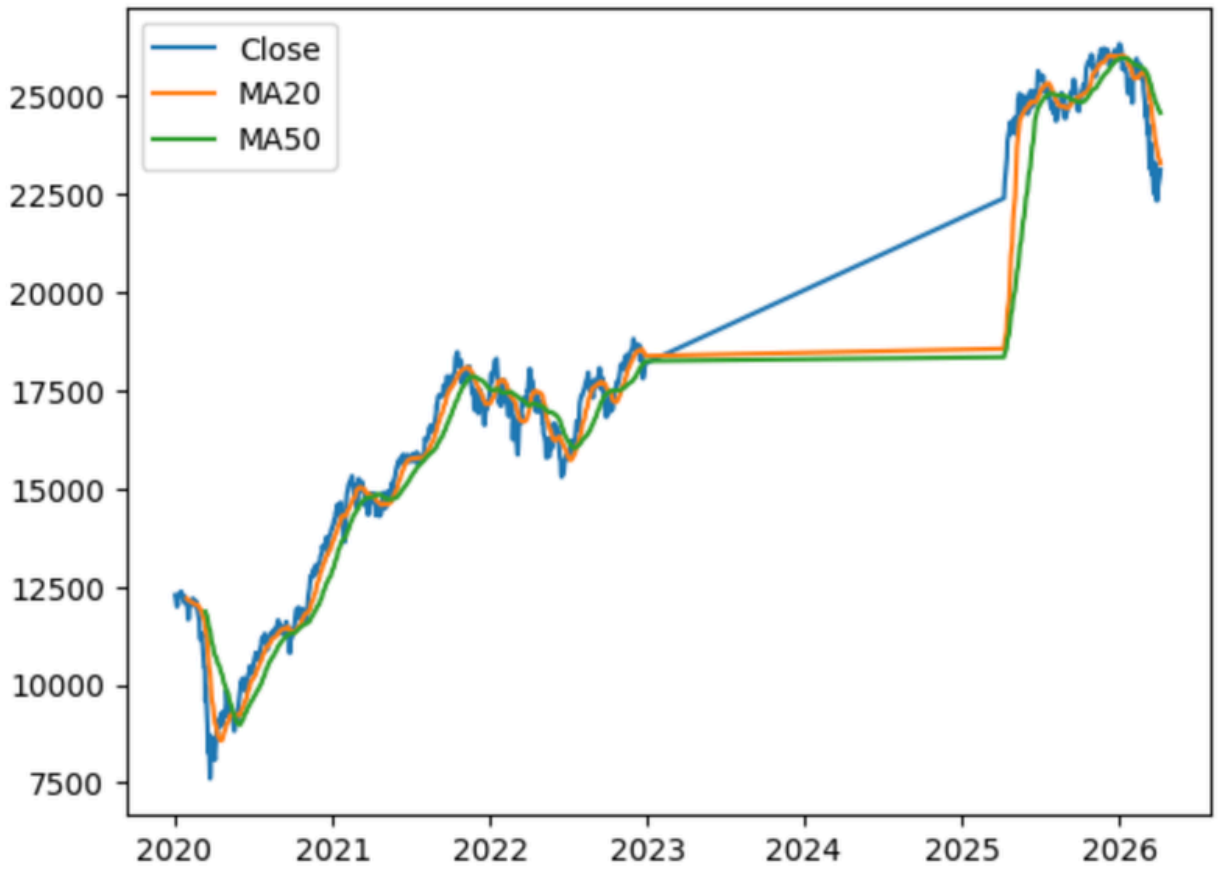
```
plt.figure()
plt.plot(df['date'], df['close'])
plt.title("Stock Price Trend")
plt.show()
```



The above graph shows the variation of stock closing prices over time. It indicates an overall upward trend with some fluctuations, reflecting market volatility and growth phases.

Moving Average Analysis

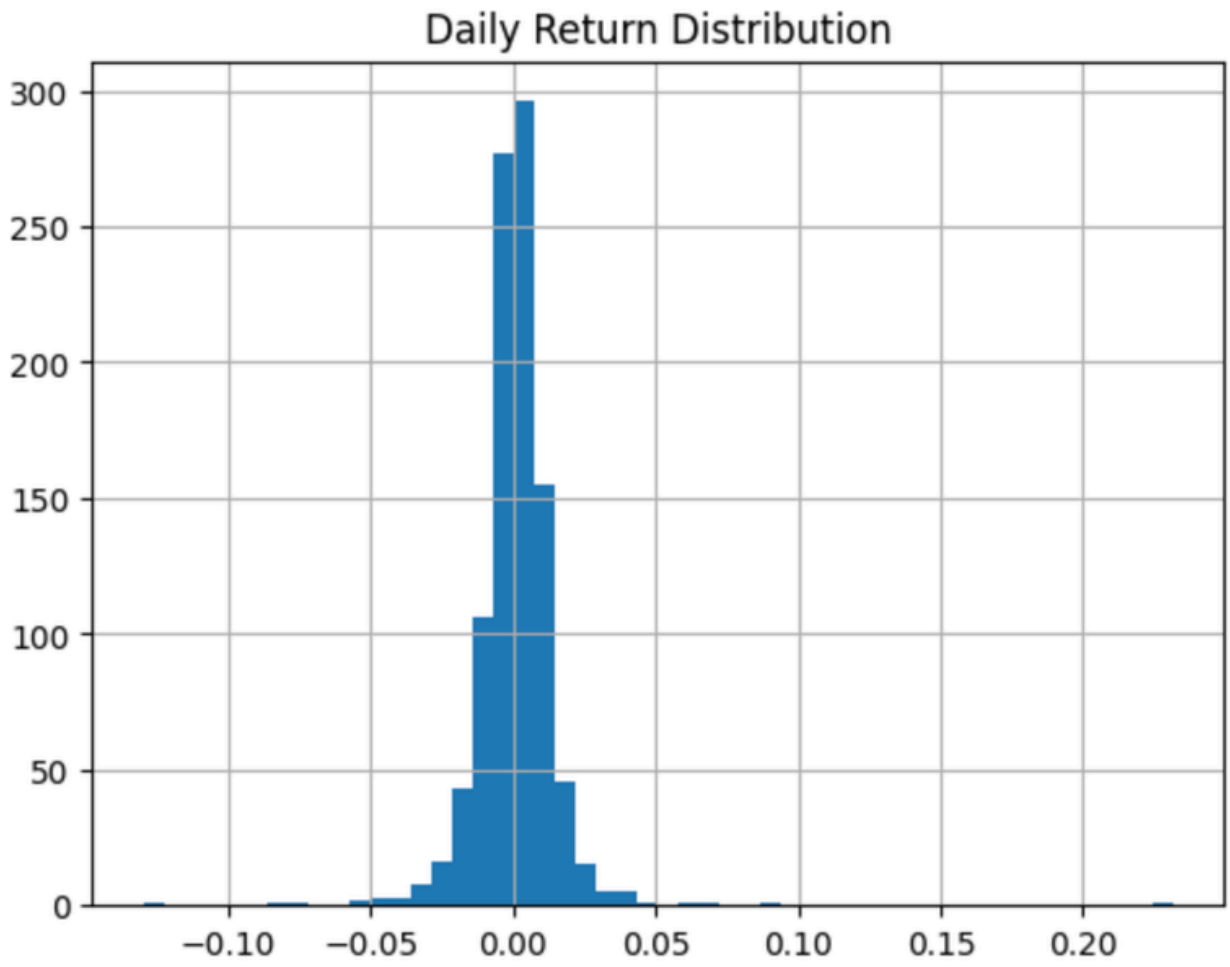
```
plt.figure()
plt.plot(df['date'], df['close'], label='Close')
plt.plot(df['date'], df['ma20'], label='MA20')
plt.plot(df['date'], df['ma50'], label='MA50')
plt.legend()
plt.show()
```



The moving average lines smooth out short-term fluctuations and highlight long-term trends. The MA20 reacts faster, while MA50 shows overall trend direction.

Daily Return Distribution

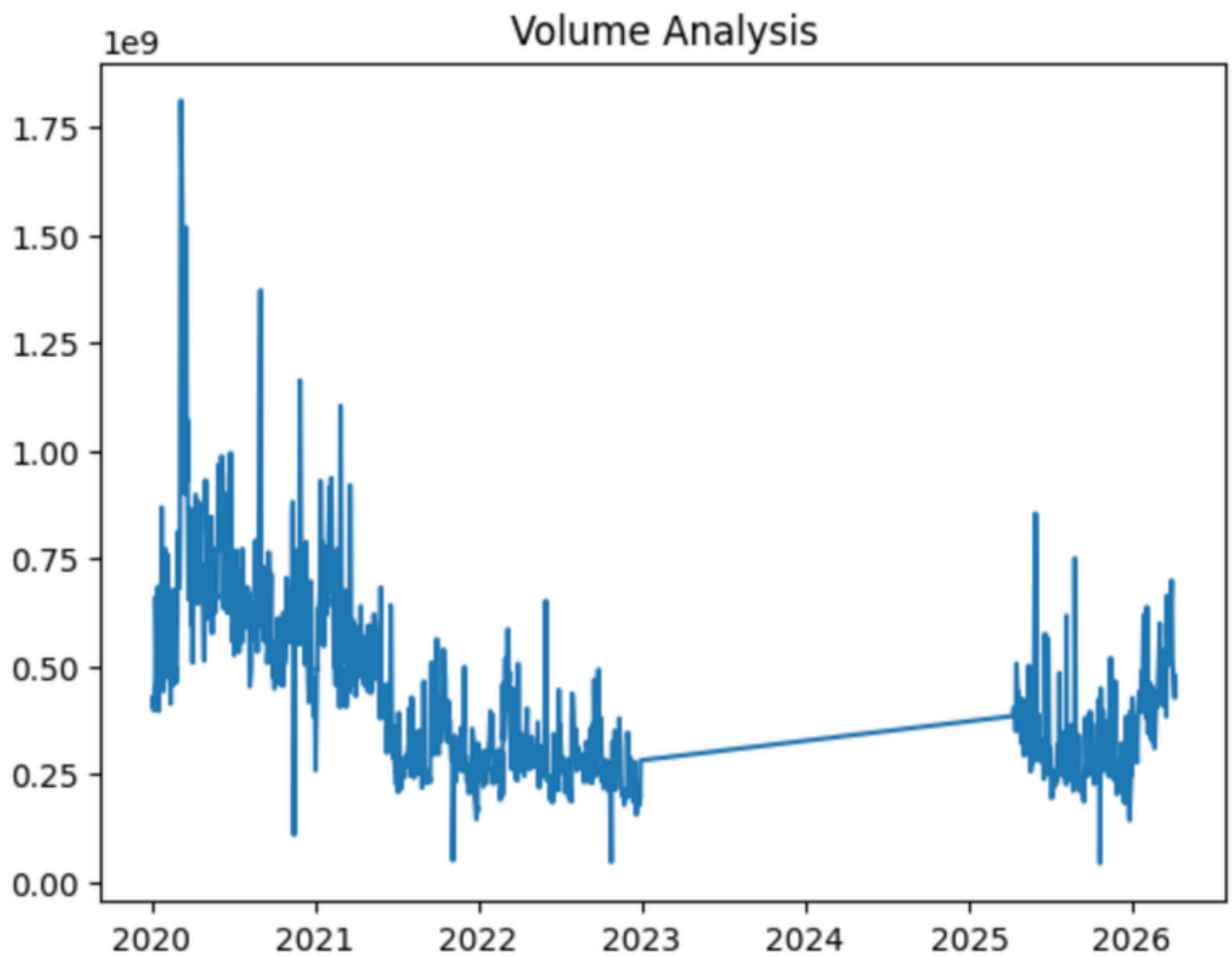
```
plt.figure()
df['daily return'].hist(bins=50)
plt.title("Daily Return Distribution")
plt.show()
```



The category-wise analysis provides a clear understanding of how different product categories contribute to overall sales in terms of quantity. It is observed that Clothing dominates the dataset with approximately 63% of total quantity sold, making it the most popular category among customers. This high share indicates frequent purchases, wider product variety, and strong customer The histogram shows the distribution of daily returns. Most values are concentrated around zero, indicating small daily changes with occasional extreme movements.

Volume analysis

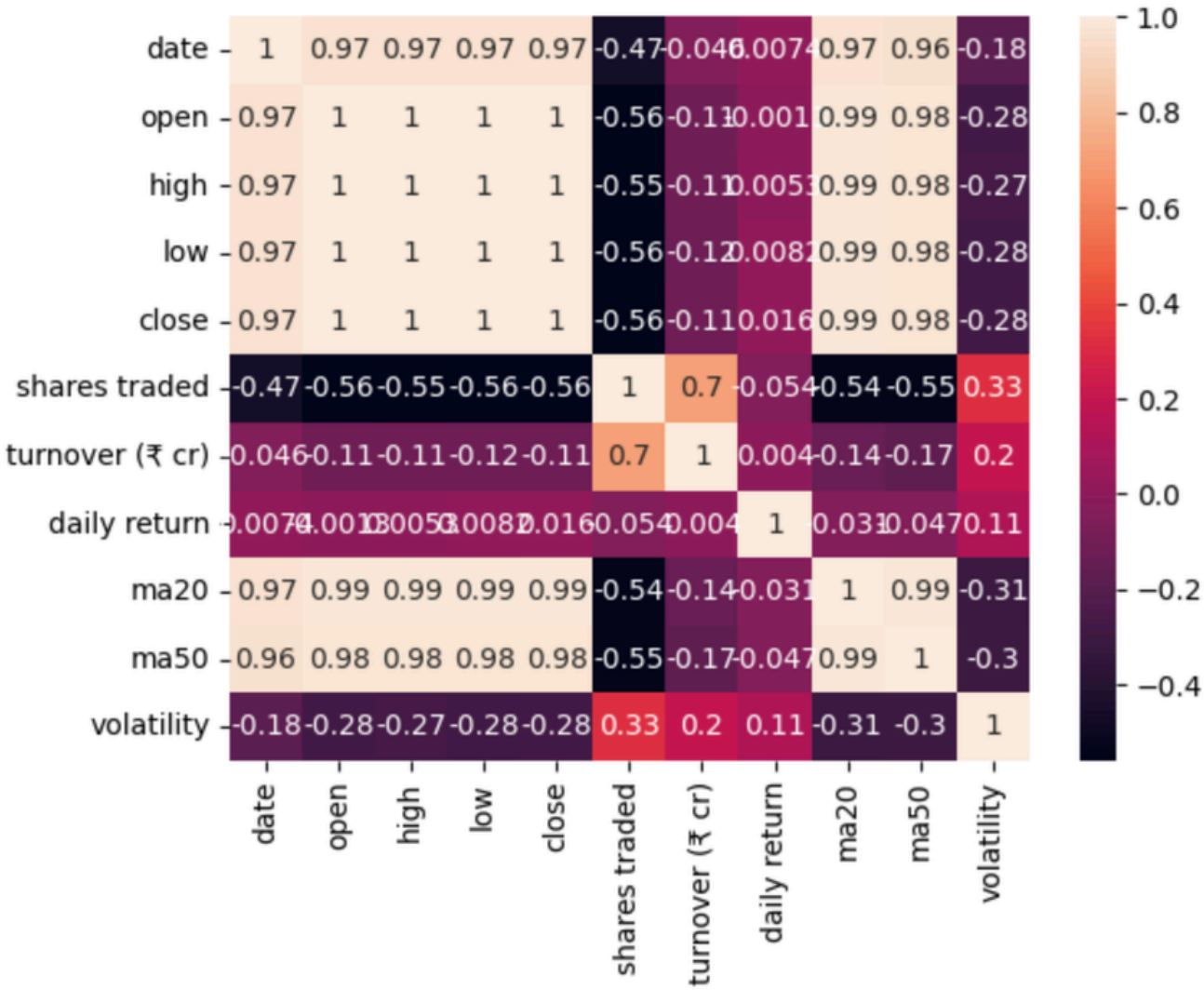
```
plt.figure()
plt.plot(df['date'],
df['shares traded'])
plt.title("Volume Analysis")
plt.show()
```



This graph represents trading volume over time. High spikes indicate increased trading activity, often linked to market events or news.

Correlation Heatmap

```
plt.figure()
sns.heatmap(df.corr(),
annot=True)
plt.show()
```



The heatmap shows relationships between variables. Strong correlation is observed between open, high, low, and close prices, while volume shows weaker correlation.

Data Understanding

df.head()
df.shape
df.info()
df.describe()

The dataset contains 989 rows and 7 columns with no missing values. It includes stock price attributes such as open, high, low, close, volume, and turnover.

7.Conclusion

- The project successfully applied Exploratory Data Analysis (EDA) on stock market data.
- Key insights about stock trends, volatility, and trading volume were identified.
- Visualizations like line plots, histograms, and heatmaps helped in understanding data patterns.
- Strong relationships were observed between stock price variables (open, high, low, close).
- The stock market shows fluctuations, making prediction difficult.
- Overall, EDA proved useful for analyzing and interpreting financial data.

8.Future Scope

- Apply advanced models like Linear Regression, ARIMA, and LSTM for prediction.
- Integrate real-time stock data using APIs.
- Develop interactive dashboards using Power BI or Streamlit.
- Analyze multiple stocks for comparative study.
- Include external factors like news, economy, and global events.
- Build a complete stock prediction and decision-support system.

9. References

- Yahoo Finance, “Stock Market Data,” Available: https://finance.yahoo.com/
- Kaggle, “Stock Market Datasets,” Available: https://www.kaggle.com/
- Scikit-learn Documentation, Available: https://scikit-learn.org/
- Pandas Documentation, Available: https://pandas.pydata.org/
- Matplotlib Documentation, Available: https://matplotlib.org/
- Seaborn Documentation, Available: Apply advanced models like Linear Regression, ARIMA, and LSTM for prediction.
- linkedIn, Available: https://www.linkedin.com/posts/hrishikesh-s-8739a0403_datascience-python-eda-ugcPost-7450423676576927744-Vdzo?utm_source=social_share_send&utm_medium=member_desktop_web&rcm=ACoAAGcyCtcBMEtX24o6hZ5-W-Q3BtZFxr7FWcs: